

4. In each of the following questions, select the correct answer.

(3 pts. each)

- (a) If A and B are 3×3 symmetric matrices, then
- (A) AB is symmetric
 - (B) AB is skew-symmetric
 - (C) $2A - 3B$ is symmetric
 - (D) None of the above
- (b) Let A and B be $n \times n$ matrices, then AB is symmetric if and only if,
- (A) A and B are symmetric
 - (B) A and B are skew-symmetric
 - (C) $(AB)^2 = A^2B^2$
 - (D) None of the above
- (c) Let A be an $m \times n$ matrix. If the linear system $AX = O_{m \times 1}$ has only the trivial solution, then for any $m \times 1$ matrix B , the system $AX = B$
- (A) has a unique solution
 - (B) has infinitely many solutions
 - (C) has no solution.
 - (D) None of the above
- (d) Let A and B be $n \times n$ diagonal matrices such that $\det(A) = \det(B)$ and $\text{tr}(A) = \text{tr}(B)$, then
- (A) $A = B$
 - (B) $\text{tr}(A^2) = \text{tr}(B^2)$
 - (C) $\det(AB) = \det(BA)$
 - (D) All of the above
- (e) Let A, B and C be $n \times n$ invertible matrices. Thus,
- (A) if $A(B + C) = C - A$, then $B = (A^{-1} - C^{-1} - I_n)C$
 - (B) $(AB)^T(AB)^{-1} = B^{-1}A^{-1}B^TA^T$
 - (C) $CA + AB = A(B + C)$
 - (D) All of the above

1. (a) (i) (5 pts.) $A^{-1} = \begin{bmatrix} -1 & 1 \\ -3 & 2 \end{bmatrix}$.
(ii) (5 pts.) $\text{tr}(A) = 1$ & $\text{tr}(A^{-1}) = 1 \implies \text{tr}(A) - \text{tr}(A^{-1}) = 0$.
(b) (5 pts.) Let $A = [a_{ij}]$, $B = [b_{ij}]$, $D = [d_{ij}] = A + cB$. Thus, $\text{tr}(A + cB) = \text{tr}(D) = \sum_{i=1}^n d_{ii} = \sum_{i=1}^n (a_{ii} + cb_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n cb_{ii} = \sum_{i=1}^n a_{ii} + c \sum_{i=1}^n b_{ii} = \text{tr}(A) + c \text{tr}(B)$.
(c) (5+5+5 pts.) $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 5 & 3 & 1 \\ 1 & 2 & -a & a \end{array} \right] \equiv \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & -a-1 & a-2 \end{array} \right]$
(i) no solution: $a = -1$ (ii) unique solution: $a \neq -1$ (iii) infinitely many solutions: Never.
2. (a) (5 pts.) (I) If A is invertible, then A^{-1} exists such that $AA^{-1} = I_n \implies \det(AA^{-1}) = \det(I_n) \implies \det(A)\det(A^{-1}) = 1 \implies \det(A) \neq 0$
(5 pts.) (II) Let $\det(A) \neq 0$. Since $\text{adj}(A)A = \det(A)I_n \implies \left[\frac{1}{\det(A)} \text{adj}(A) \right] A = I_n \implies A^{-1} = \frac{1}{\det(A)} \text{adj}(A) \implies A$ is invertible.
(b) (10 pts.) Let A be a nonsingular matrix and suppose that X and Y are inverses of A . Thus, $AX = I = XA$ and $AY = I = YA$. Now, $X = XI = X(AY) = (XA)Y = IY = Y$.
(c) (10 pts.) Let $S = \frac{1}{2}(A + A^T)$, $K = \frac{1}{2}(A - A^T)$. Clearly, $S + K = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = A$ and $S^T = \left[\frac{1}{2}(A + A^T) \right]^T = \frac{1}{2}(A^T + (A^T)^T) = S$ & $K^T = \left[\frac{1}{2}(A - A^T) \right]^T = \frac{1}{2}(A^T - A) = -K$.
3. (a) (10 pts.) $\det(\text{adj}(2B)) = [\det(2B)]^2 = [2^3 \det(B)]^2 = [2^3(-\frac{1}{2})]^2 = 16$.
(b) (i) (5 pts.) $\det(A^{-1}) = \begin{vmatrix} 1 & -2 & 2 & 1 \\ 1 & 1 & -1 & 4 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 2 & 1 \\ 0 & 3 & -3 & 3 \\ 0 & -1 & 3 & 2 \\ 0 & -1 & 1 & 3 \end{vmatrix} = 3 \begin{vmatrix} 1 & -1 & 1 \\ -1 & 3 & 2 \\ -1 & 1 & 3 \end{vmatrix} =$
 $3 \begin{vmatrix} 1 & -1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{vmatrix} = 24 \implies \det(A) = \frac{1}{24}$.
(ii) (5 pts.) $\text{adj}(A) = \det(A)A^{-1} = \frac{1}{24} \begin{bmatrix} 1 & -2 & 2 & 1 \\ 1 & 1 & -1 & 4 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{24} & -\frac{1}{12} & \frac{1}{12} & \frac{1}{24} \\ \frac{1}{24} & \frac{1}{24} & -\frac{1}{24} & \frac{1}{6} \\ -\frac{1}{24} & \frac{1}{24} & \frac{1}{24} & \frac{1}{24} \\ -\frac{1}{24} & \frac{1}{24} & -\frac{1}{24} & \frac{1}{12} \end{bmatrix}$.
(iii) (1 point) The solution of $AX = O$ is the trivial solution since A is invertible.
[i.e., $x_1 = x_2 = x_3 = x_4 = 0$].
(4 pts.) The solution of $AX = \begin{bmatrix} 1 & 0 & 0 & -1 \end{bmatrix}^T$ is $X = A^{-1} \begin{bmatrix} 1 & 0 & 0 & -1 \end{bmatrix}^T \implies x_1 = 0, x_2 = -3, x_3 = -2, x_4 = -3$.

4. (a) If A and B are 3×3 symmetric matrices, then
 (C) $2A - 3B$ is symmetric
- (b) Let A and B be $n \times n$ matrices, then AB is symmetric if and only if,
 (D) None of the above
- (c) Let A be an $m \times n$ matrix. If the linear system $AX = O_{m \times 1}$ has only the trivial solution, then for any $m \times 1$ matrix B , the system $AX = B$
 (D) None of the above
- (d) Let A and B be $n \times n$ diagonal matrices such that $\det(A) = \det(B)$ and $\text{tr}(A) = \text{tr}(B)$, then
 (C) $\det(AB) = \det(BA)$
- (e) Let A, B and C be $n \times n$ invertible matrices. Thus,
 (A) if $A(B + C) = C - A$, then $B = (A^{-1} - C^{-1} - I_n)C$